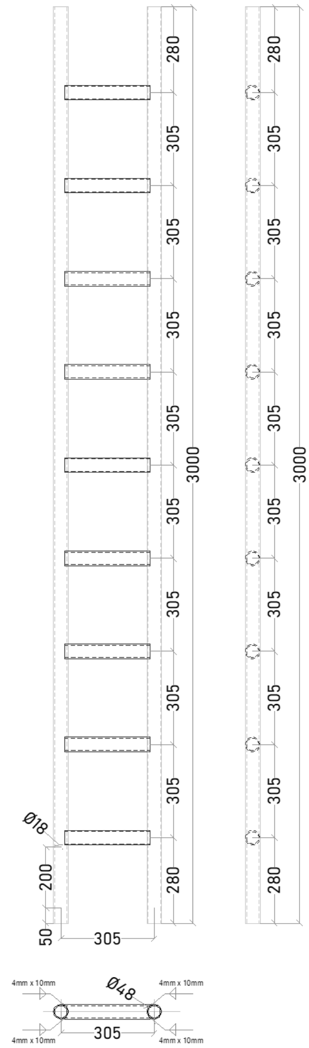
Data Sheet for weld Resistance at Steel Ladder

- Welding Data:

Weld Thickness = 4.0 mm
Length = 20 mm (at one side of stair beam)
Steel Grade: S235 (fu = 360 (N/mm²))
Partial safety factor for welds (γM2): 1.25
Weld Type: fillet weld



- Design shear strength of weld metal:

$$f_{vw,d} = \frac{f_u}{\sqrt{3} \cdot \gamma_{M2}}$$

- $f_u = 360$ N/mm^2 (ultimate tensile strength of base steel)
- $\gamma_{M2} = 1.25$ (partial safety factor for welds)

$$f_{vw,d} = \frac{360}{\sqrt{3} \times 1.25} = \frac{360}{2.165} = 166.3 \text{ N/mm}^2$$

- Effective Weld Area:

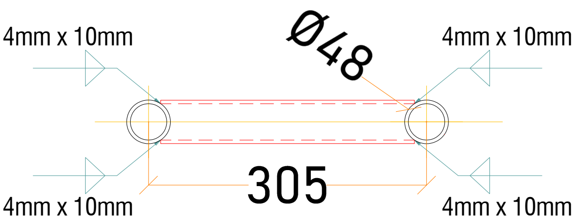
Throat thickness (a): 4.0 mm
Weld length (l_w): 20 mm

- Effective area per Side: A = a × l_w = 4.0 × 20 = 80 mm²

- Weld Resistance:

- $F_{w,Rd}$ = A× $f_{vw,d}$ = 80 × 166.3 = 13304 N = 13.30 kN

- **Weld Resistance for two sided of Stair beam= 13.30x2= 26.60 kN**



Head Office
Factory: Wady Houf, Helwan
Tel: (+202) 23254532 / 23254519
(+20223254534
Fax: (+202) 23671 347
E-mail: sales@acrow.co